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AN ON-LINE DIAGNOSTIC WIND MODEL APPLIED TO THE SAN FRANCISCO BAY REGION

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1. INTRODUCTION

The result of the work described here can be seen on the World Wide Web at the following address:

<http://sfbay7.wr.usgs.gov/~jonathan/wind.html>

The wind display that is found at that site is the result of a cooperative effort among many individuals and organizations. The data that are used are contributed to a common database hourly by numerous organizations, private and at several levels of government. The data assimilation is done in three steps: 1) Reading and decoding, 2) calculation of the winds, and 3) the display of the results. The remainder of this paper describes the technical aspects of the interactive wind display, including the origins of the data and how they are translated, a description of the diagnostic wind model that is used, and the features of the web page. As always, improvements are possible, and some of those are discussed at the end of the paper.

2. THE DATA

The model will be discussed briefly in the next section. Here, it is sufficient to note that its inputs are wind data from surface sites, and whatever wind and temperature profiles are available. Unfortunately, we are currently using upper level measurements from only one location, Oakland International Airport, where a radiosonde is launched twice daily. Thus the model is forced to use the most recent upper air observations, which can be over twelve hours old.

For the most part, the surface observations are made hourly, although some coastal buoys report at three hour intervals. The inputs include the standard observations from the National Weather Service (NWS), FAA, NOAA, Navy, Coast Guard, and the Air Force. The Bay Area Air Quality Management District (BAAQMD) contributes data from their network of stations, as do several private and semi-private organizations, such as Pacific Gas and Electric (PG&E), Lawrence Livermore National Laboratory (LLNL), Chevron, and Union Oil.

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Currently, data are available from more than 100 sites in Northern and Central California. Figure 1 shows the locations of those sites that are within and around the calculation domain. There are about 60 sites within the immediate San Francisco Bay region for which the winds are displayed. Not all stations are available every hour, but there are usually thirty or more stations from which recent wind observations are available.

The collection and processing of the data involves a rather circuitous procedure. The NWS automatically polls and collects observations from many of the sites hourly and updates the Automated Local Event Reporting in Real-Time (ALERT) file. A 486 DX4 PC at the Meteorology Department at San Jose State University (SJSU) polls the ALERT file and PG&E and BAAQMD stations via the phone system and a modem. The raw data are received in a form that includes a large amount of text that must be processed to produce formatted files for input to the diagnostic wind model, so the raw data files are then transmitted over an ethernet connection to a workstation at SJSU, and from there to a workstation at the Center Weather Service Unit (CWSU) in Fremont (CA) over a dedicated digital phone line. The CWSU workstation ftp's raw data to a PC that "parses" the raw data and creates files used by the WOCSS model. The parsing PC runs a version of the WOCSS model for display at CWSU (Strach et al., 1997), and also transmits the input data files back to SJSU workstation, which then ftp's the parsed data to a web-server workstation at the U. S. Geological Survey's (USGS) Western Regional Office in Menlo Park via an internet connection.

3. THE WIND ANALYSIS MODEL

The hourly observations are used to generate wind fields using the Winds on Critical Streamline Surfaces (WOCSS) approach described in detail by Ludwig et al. (1991). This is an interpolation scheme that is guided by the critical dividing streamline concept (e.g. McNider, et al., 1984) in its treatment of terrain induced vertical motion, and by mass conservation constraints. The WOCSS methodology evolved from the wind-energy planning model of Bhumraikar et al. (1980) which applied a variational calculus numerical scheme similar to Sherman's (1978), except that they used "sigma" coordinates rather than the Cartesian coordinates used by Sherman. Endlich (1984) introduced an iterative technique (Endlich, 1967) for removing divergence, and

allowed the coordinate surfaces to intersect the terrain to force the flow around the intersected obstacles when the nondivergence constraint was invoked. Endlich (1984) defined coordinate surface shapes subjectively. The wind calculation scheme employed on the web page uses the results of field and laboratory studies (e.g. Hunt and Snyder, 1980; Lavery et al., 1982) that generally verified Sheppard's (1956) analysis of how the meteorological conditions cause a flow to pass over an obstacle, rather than around it. As described below, Ludwig et al. (1991) adapted these studies to define flow surface shapes during stable atmospheric conditions.

When potential temperature increases with height and atmospheric processes are approximately adiabatic, there will be a height where the work done in displacing the air from its equilibrium position against the buoyant restoring force equals its original kinetic energy. According to the critical streamline concept, this defines the maximum obstacle height that can be surmounted by the flow. This equivalence of potential and kinetic energies can be expressed in terms of Z_{max} , the greatest height to which the air at height Z_0 can be lifted by the local wind speed (V_0) against the local potential temperature gradient, $d\theta/dz$:

$$Z_{max} - Z_0 = V_0 \left(\frac{g}{\bar{T}} \frac{d\theta}{dz} \right)^{-\frac{1}{2}} \quad (1)$$

$\bar{T}$ is the average temperature in the layer between Z_0 and Z_{max} . Equation (1) is used to define coordinate surfaces that approximate the shape of the flow,

intersecting the terrain in areas where the flow cannot pass over it. Winds are set to zero at points below the local terrain, so adjustments toward nondivergence cause the flow to pass around obstacles.

Use of surfaces defined by Equation (1) not only incorporates the critical dividing streamline effects into the analysis, but also reduces the three-dimensional problem to several, more easily solved two-dimensional problems. The steps involved in the overall wind calculation process are as follows:

1. Find the lowest terrain.
2. Specify flow surface heights over lowest terrain
3. Specify a set of terrain-following surfaces that have the specified heights over the low terrain.
4. Interpolate the observed winds to grid points on these terrain-following surfaces.
5. Estimate the maximum height to which the air can rise from its height over the low terrain using Equation 1, and the observed temperature and wind profiles.
6. Constrain maximum rise to prevent a layer from rising more than the next lower layer, or from approaching too close to the next higher layer.
7. Define flow surfaces so that the maximum rise is over the highest terrain, and is proportional to elevation differences elsewhere.

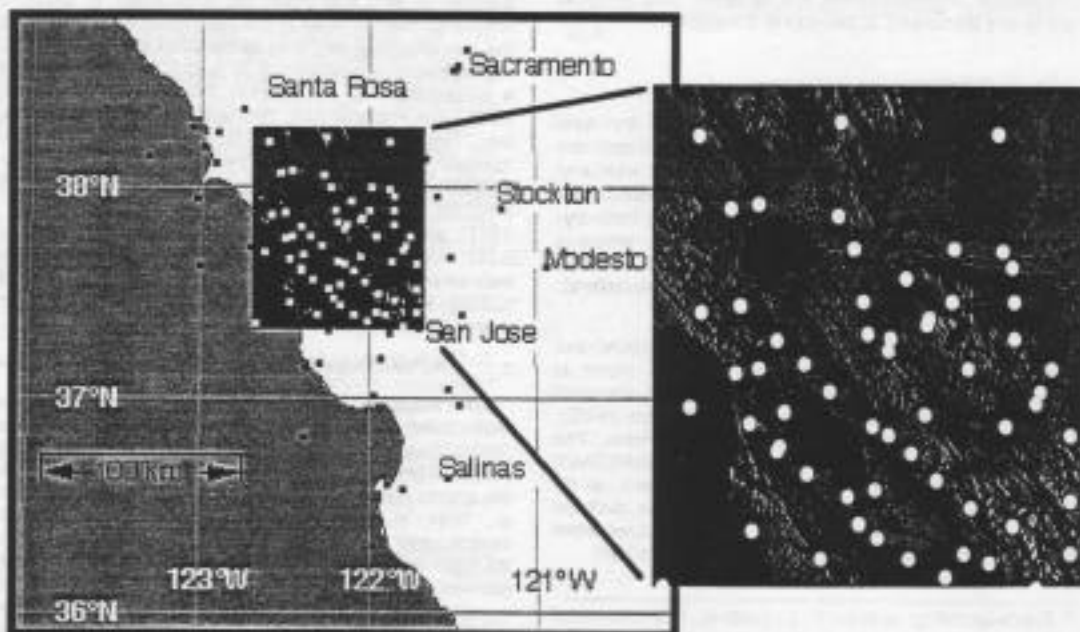


FIGURE 1 LOCATIONS OF STATIONS, AND THE COMPUTATION DOMAIN USED FOR WIND FIELD GENERATION

8. Reinterpolate winds to the new flow surfaces.
9. Assign zero winds where a surfaces intersect the terrain.
10. Adjust the flow on the surfaces toward two-dimensional nondivergence; this will force flow around terrain obstacles.
11. Interpolate the winds on the flow surfaces back to constant altitude surfaces, or to terrain-following surfaces.

The winds displayed on the web page are based on calculations made at 6 levels on a 106x123 grid covering the domain shown by the inset in Figure 1. The horizontal grid spacing is 1 km. The calculations take approximately 10 to 20 second of cpu on a typical workstation. Results are interpolated to anemometer height (10 m) and displayed for every third grid point, i. e. the grid spacing for the displayed winds is 3 km.

4. WEB PAGE DESIGN

The web page described here was not the primary reason for the work that has been done. The original wind model development was for field use by the Army. The data gathering and processing was primarily to meet needs for timely information that could be used for air traffic control purposes in the San Francisco Bay Area (Strach et al., 1997), and the detailed analyses by the USGS are to provide wind stress inputs for models of currents in the San Francisco Bay and San Joaquin River Delta. However, the world wide web provides a convenient method for disseminating the results to others who might find them useful.



FIGURE 2 EXAMPLE OF WIND FIELD DISPLAY

The main feature of the page is the display of the wind field over the San Francisco Bay area. An example of this display is shown in Figure 2. The original display is in color, with wind speeds indicated both by the size of the vector, and by its color. The wind field is updated hourly, usually about 30 minutes after the observations are made. Thus, the wind field will usually represent conditions about 30 to 90 minutes earlier.

The winds are calculated on a finer grid than is displayed, so the user is given the option of moving the cursor to any particular point of interest, clicking and getting the wind speed and direction for that point. These winds are not just interpolations between the vectors that are displayed; they make use of the calculations on the finer grid, and hence reflect the effects of more detailed topography.

The page also includes an option for retrieving wind fields for any of the preceding 23 hours, or an animated sequence of wind images for those hours. These recent wind fields are stored and delivered as GIF imagery. At the time of writing, a more comprehensive archiving system was also being tested. With that system, the user can select days and hours for the past several months and the program will calculate and display the wind fields.

5. USES AND POSSIBLE IMPROVEMENTS

The assumption that the wind fields that are generated would be of interest to others has proved justified. More than 80 thousand "hits" were recorded during the first six months that the site was operational, and it is referenced as a link by more than 60 other sites. Users include construction companies, wind surfers, hang glider and sail plane pilots, and sailors, as well as the expected meteorological organizations (e.g. the local NWS Office, TV Weather broadcasters and meteorology departments).

Some users have made suggestions for improving the display. The most common of these is that the observed winds be displayed. This should not be difficult to do, and probably will be done in the future. Another common suggestion is that more information should be provided about how the model works, and the origins of the data. This preprint provides a basis for satisfying that suggestion, and we expect to adapt it and link it to the page.

As noted earlier, only the Oakland sounding is available as a source of upper air information. There are other, better sources now operating in the region. These include profilers that measure winds and temperatures aloft on a nearly continuous basis at Bodega Bay and Fort Ord. These sites are just north and south of the region, respectively. Sodars are also operating at the San Francisco airport (SFO), on the Pacific coast to the west of SFO at Half Moon Bay, and on the Bay to the south of SFO at San Carlos. Sodar data are not directly usable, but they can provide information that could be used to adjust the most recent sounding information to provide estimates of the current

inversion height. There are no plans at this time to use these other data sources.

There have also been questions about establishing similar pages for other regions, such as the Chesapeake Bay. The basic wind calculation code is available (for a diskette and postage) from the corresponding author. That FORTRAN code has proven fairly easy to adapt to different machines. It has been run in several parts of the world on PCs, Macintoshes, VAX systems and UNIX machines. The greatest difficulty is likely to be the acquisition and processing of current data.

The archiving of data in connection with the web page and the air traffic uses (Strach et al., 1997) of the wind code have provided the means for a more thorough evaluation of its performance than has been done in the past (e. g., Bridger et al., 1994; Thykier-Nielsen et al., 1990). The WOCSS model can be applied to a wide variety of cases, using only a limited subset of the available input data, and the results compared with those observations that were not used. This will quantify the degree to which the WOCSS analysis scheme is able to reproduce winds in areas without observations. It should also identify those conditions for which the methodology is unreliable, which may allow the approach to be refined and improved.

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